Manav Mepani

Worcester, MA 01609 (open to relocate) | +1 (732) 880 1082 | mepanimanav@gmail.com | LinkedIn/Manav Mepani | Portfolio

EDUCATION

Worcester Polytechnic Institute (WPI), USA **Aug 2024 - May 2026**
Master of Science, Mechatronics (GPA: 4.0/4.0) *USA*

Sardar Patel College of Engineering (SPCE), India **Aug 2018 - Jun 2022**
Bachelor of Technology, Mechanical Engineering (GPA: 9.04/10) *India*

EXPERIENCE

EssilorLuxottica – GENTEX (MA, USA) | *Automation Engineer Intern* **May 2025 - Present**

- Redesigned and optimized a locking system for manufacturability, and designed and fabricated a secondary base locking mechanism for an existing assembly
- Conducted root cause analysis to improve automated label-inspection accuracy and reduced customer mispack complaints by developing and optimizing computer-vision checks using the Keyence Vision System.
- Migrated Mitsubishi robot code to Epson, refactored for clarity, redesigned the homing sequence, and integrated robot control with Allen-Bradley PLCs for synchronized line operation.
- Produced clear technical documentation and worked extensively with industrial automation systems, including Magnemotion carriers, 6-axis robots, SCARA, and gantry systems while applying mechanical engineering principles.

The Innovation Story | *Robotics Mechanical Design Engineer* **Aug 2023 - Apr 2024**

- Applied mechanical engineering principles to design and manufacture complex systems for FTC and FRC game challenges, achieving ~90% reliability within three months using innovative designs and in-house manufacturing.
- Worked with teams of 20 students of various grades while teaching them physics, CAD, and manufacturing for the competition in which they stood 4th all over India.

Technology Innovation Hub, IIT Bombay | *Project Research Assistant* **Jun 2022 - Aug 2023**

- Used SOLIDWORKS to design and manufacture an autonomous onion harvester for small farms, featuring an Energy-Aware Path Planning algorithm and a return-to-base function triggered by low battery levels to ensure operational efficiency.
- Improved the maneuverability of a bomb disposal rover by using a design that ensures at least one side's wheels maintain ground contact for enhanced stability, leveraging mechanical engineering principles.
- Optimized rocker-bogie mechanism design for a rover using Simulated Annealing optimization algorithm in Octave to enhance step-climbing ability by analyzing key design parameter relationships.

Prolific 3D Tech | *Intern Design Engineer* **Jun 2021 - Jul 2021**

- Led a reverse engineering project to replicate and modify a Can Filling and Seaming Machine, creating 3D models and 2D manufacturing drawings in SOLIDWORKS after precise part measurements.

SKILLS

- **Softwares:** CATIA V5, SOLIDWORKS, AutoCAD, ANSYS Workbench, ADAMS, FluidSIM, MS OFFICE SUIT, Studio 5000 (Allen-Bradley)
- **Mechanical Skills:** CAD Software, Mechanical Engineering Principles, GD&T, Root Cause Analysis, Engineering Documentation, Process Improvement
- **Programming Skills:** Python, C++, Octave, Arduino, Webots, Robot Operating System (ROS2), MATLAB, Epson RC+ (SPEL+), RT ToolBox3
- **Certifications:** Digital Electronic circuits, ANSYS Workbench, SIEMENS PLC

PUBLICATIONS

Energy-and Safety-Aware Operation of Battery-Powered Autonomous Robots in Agriculture.
IEEE Transactions on AgriFood Electronics (2024).

Study of the Step Climbing Ability of a Rover with Rocker Bogie Mechanism.
Journal of The Institution of Engineers (India): Series C (2024): 1-17.

Design of Robot Arm for Domestic Culinary Assistance.
Materials Today: Proceedings, Volume 68, Part 6, 2022, Pages 1930-1945, (ICAME22 Conference)

ACADEMIC PROJECTS

Design and Development of an Autonomous Cooking System **Aug 2021 - May 2022**

- Designed, fabricated and analysed the design for a 6 DOF PUMA-configured robotic arm for autonomous cooking.
- Simulated Inverse Kinematics algorithm which used kinematic decoupling method in Webots and controlled the physical model using an Arduino microcontroller.
- Achieved a 0.7 Kg (1.55 lbs) payload capacity with moderately high repeatability of ±1° error in motion control using PID.